

Last Time:

• Def: V v.s.

(1) $v \in V$ is a l.c. of $w_1, \dots, w_k \in V$ if

$$\exists a_1, \dots, a_k \in \mathbb{R} \text{ s.t. } v = a_1 w_1 + \dots + a_k w_k.$$

(2) $v \in V$ is a l.c. of S (where $\emptyset \neq S \subseteq V$) if

$$\exists \text{ fin } \emptyset \neq T = \{w_1, \dots, w_k\} \subseteq S \text{ s.t. } v \text{ is a l.c. of } w_1, \dots, w_k.$$

(3) $v \in V$ is a l.c. of $\emptyset \subseteq V$ if $v = 0$.

• Def: V v.s. can treat this as a list
 $(w_1, \dots, w_k) \in V^k := \{(v_1, \dots, v_k) : v_i \in V \ \forall i\}$

(1) $\underbrace{w_1, \dots, w_k}_{\substack{\text{can treat this as a list} \\ \nearrow}} \in V$ then

$$\text{span}(w_1, \dots, w_k) := \left\{ \sum_{i=1}^k a_i w_i : a_i \in \mathbb{R} \ \forall 1 \leq i \leq k \right\} \subseteq V.$$

set of all l.c.s of $w_1, \dots, w_k$

(2) $S \subseteq V$ then

$$\text{span}(S) := \{v \in V : v \text{ is a l.c. of } S\} \subseteq V.$$

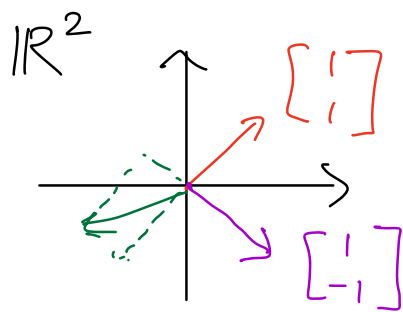
Today:

Continue §1.3 Linear Combinations, Span, and

Linear Independence (span(S) subsp of V ?)

Ex 1.32:

$$(a) \operatorname{span} \left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right) = \mathbb{R}^2.$$



Pf: Let $W := \operatorname{span} \left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right) \subseteq \mathbb{R}^2$ by def.

WTS: $\forall \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2, \begin{bmatrix} x \\ y \end{bmatrix} \in W.$

$$W = \left\{ a \begin{bmatrix} 1 \\ 1 \end{bmatrix} + b \begin{bmatrix} 1 \\ -1 \end{bmatrix} : a, b \in \mathbb{R} \right\},$$

$$\begin{bmatrix} a+b \\ a-b \end{bmatrix}$$

so need to solve a & b for

$$\begin{bmatrix} a+b \\ a-b \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} \Leftrightarrow \begin{cases} a+b = x & (1) \\ a-b = y & (2) \end{cases} \left. \vphantom{\begin{matrix} a+b = x \\ a-b = y \end{matrix}} \right\} \begin{array}{l} \text{system of} \\ \text{linear equations} \end{array}$$

Let's try the method of elimination:

$$(1) + (2): 2a = x + y \Rightarrow a = \frac{x+y}{2}$$

substitution

$$(1): \frac{x+y}{2} + b = x \Rightarrow b = \frac{x}{1} - \frac{x+y}{2} = \frac{x-y}{2} \quad \square$$

soln
solution
to
(1) & (2)

$$(b) \operatorname{span} \left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right) = \left\{ a \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} : a, b \in \mathbb{R} \right\}$$

$$= \left\{ \begin{bmatrix} a \\ b \\ 0 \end{bmatrix} : a, b \in \mathbb{R} \right\} \neq \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

xy -plane

$$\Rightarrow \text{span} \left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right) \subsetneq \mathbb{R}^3$$

$$(c) \quad 2 + 3x + x^2 \in \text{span}(1, 1+x, (1+x)^2) \subseteq \mathbb{R}[x]_{\leq 2} \quad (?)$$

Sol: Solve a, b, c for

$$2 + 3x + x^2 = a \cdot 1 + b(1+x) + c(1+x)^2$$

$$= \underbrace{a+b+c} + \underbrace{bx} + \underbrace{c} + \underbrace{2cx} + \underbrace{cx^2}$$

$$= \underbrace{(a+b+c)} + \underbrace{(b+2c)x} + \underbrace{cx^2}$$

compare coeffs $\swarrow$

$$(1) \quad a+b+c=2 \quad \Rightarrow (2): b+2 \cdot 1=3$$

$$(2) \quad b+2c=3 \quad \Rightarrow \underline{b=1}$$

$$(3) \quad \underline{c=1} \quad \Rightarrow (1): a+1+1=2 \Rightarrow \underline{a=0}$$

Thus, $2 + 3x + x^2 = 0 \cdot 1 + (1+x) + (1+x)^2$ is a

l.c. of $1, 1+x, (1+x)^2$. So YES!

In Ex 1.32(a)(b), note that spans are subsp.

Is $\text{span}(S) \subseteq V$ always a subsp?

$\Downarrow$ Yes

Prop 1.33: V v.s. Then $\forall S \subseteq V$, $\text{span}(S)$ is the smallest subsp of V containing S . (In other i.e. words, if $W \subseteq V$ subsp and $S \subseteq W$, then $\text{span}(S) \subseteq W$)

subsp and $\text{span}(S) \subseteq W$

Pf: See FIS Thm 1.5 (uses Subsp Crit). $\square$

Rmk: V v.s.

(1) If $v_1, \dots, v_n \in V$, then

by Prop 1.33

$$\text{span}(v_1, \dots, v_n) \stackrel{\text{by def}}{=} \text{span}(\{v_i : 1 \leq i \leq n\}) \subseteq V \text{ subsp}$$

by def

$= \{v_1, \dots, v_n\}$ if $v_i \neq v_j \forall i \neq j$
(otherwise, remove repetitions)

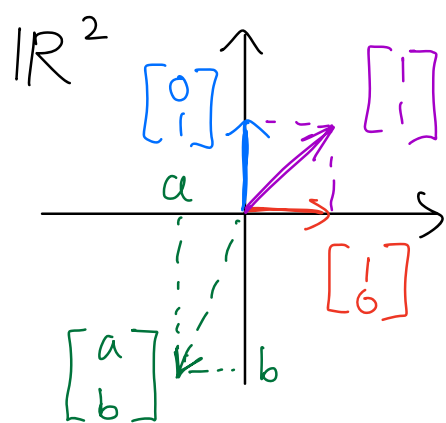
(2) $S \subseteq V$. Then Prop 1.33 and Rmk 1.29(2)

$$\Rightarrow \text{span}(S) = \bigcap_{\substack{W \subseteq V \text{ subsp} \\ S \subseteq W}} W.$$

Def 1.34: V v.s., $W \subseteq V$ subsp. A spanning set of W is a subset $S \subseteq W$ s.t. $W = \text{span}(S)$. In this case, we say S spans (or generates) W .

Linear (In)dependence

Motivation: Consider $\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix} \in \mathbb{R}^2$.



$$\begin{bmatrix} a \\ b \end{bmatrix} = \underbrace{a \begin{bmatrix} 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \end{bmatrix}}_{\text{l.c. of } \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}} + 0 \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

l.c. of $\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$$\text{So } \mathbb{R}^2 \supseteq \text{span} \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right) \supseteq \text{span} \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) = \mathbb{R}^2$$

$\Rightarrow$ adding $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ to $\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ does not grow subsp
(so "redundant" !)

Q: How to tell when $S \subseteq V$ is "redundant"?

Let's experiment:

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} = 1 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 1 \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} + 0 \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad (1) \quad \left. \begin{array}{l} \end{array} \right\} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{ as l.c. of}$$

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} = 0 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 0 \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} + 1 \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad (2) \quad \left. \begin{array}{l} \end{array} \right\} \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{array}{l} (1) - (2) \\ \Rightarrow 0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 1 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 1 \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} - 1 \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} \end{array}$$

$$\underline{0 = \text{l.c. of } \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{ w/ non zero coeffs}}$$

↑

Impossible for 0 as a l.c. of $\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ b/c

$$0 = a \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + b \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} \Rightarrow 0 = \begin{bmatrix} a \\ b \end{bmatrix} \Rightarrow a = b = 0.$$

$\Downarrow$ benchmark this experiment

Def 1.35: V v.s.

(1) $v_1, \dots, v_n \in V$ is linearly ^{l.d.} dependent if

$$\exists (a_1, \dots, a_n) \in \mathbb{R}^n \setminus \{0\} \text{ s.t. } \sum_{i=1}^n a_i v_i = 0.$$

means $\exists j$ s.t. $a_j \neq 0$ $(X \setminus Y := \{x \in X : x \notin Y\})$

(holds when $a_1 = \dots = a_n = 0$, so we are skipping that case)

(2) Subset $S \subseteq V$ is l.d. if $\exists T = \{v_1, \dots, v_k\} \subseteq S$ s.t. $T \neq \emptyset$ (so $k \geq 1$) and $v_1, \dots, v_k$ is l.d. as in (1).

Note: Similar to span, if $S = \{v_1, \dots, v_n\} \subseteq V$, then S l.d. if and only if $v_1, \dots, v_n$ l.d.

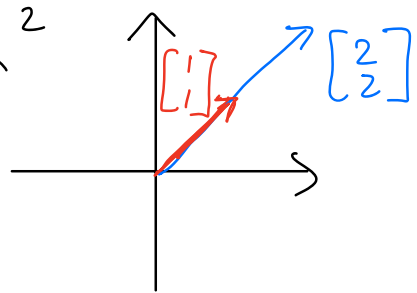
Ex 1.36: V v.s., $v_1, \dots, v_n \in V$.

(1) $\exists i \neq j$ s.t. $v_i = v_j \Rightarrow v_1, \dots, v_n$ l.d.

$$\left(\begin{array}{l} \text{b/c } 0 = 1 \cdot v_i - 1 \cdot v_j = \sum_{l=1}^n a_l v_l \text{ where} \\ a_l = \begin{cases} 0 & \text{if } l \neq i, j \\ 1 & \text{if } l = i \\ -1 & \text{if } l = j \end{cases} \end{array} \right)$$

(2) $\exists i$ s.t. $v_i = 0$, then $v_1, \dots, v_n$ is l.d., b/c

$0 = 1 \cdot v_i$ a l.c. of $v_1, \dots, v_n$ w/ a nonzero coeff.

(3) $\mathbb{R}^2$  Then $\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \end{bmatrix}$ is l.d. b/c
 $0 = 2 \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} - 1 \cdot \begin{bmatrix} 2 \\ 2 \end{bmatrix}$

$\Downarrow$ generalization

Exercise: If $v_2 = cv_1$ for some $c \in \mathbb{R}$, then v_1, v_2 is l.d.

(4) From Motivation, $\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ l.d. but
 $\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ is not l.d.
 $\Downarrow$

Def 1.37: V v.s.

(1) $v_1, \dots, v_n \in V$ is linearly independent ^{l.i.} if it's not l.d.

(2) $S \subseteq V$ is l.i. if not l.d.

(By def, $\emptyset$ is not l.d. $\Rightarrow \emptyset$ is l.i.)

Ex 1.38:

(1) In $\mathbb{R}^n$, $\forall 1 \leq i \leq n$, let $e_i \in \mathbb{R}^n$ be a col

vec s.t. $(e_i)_j = \begin{cases} 1 & \text{if } j = i \\ 0 & \text{if } j \neq i \end{cases}$ (e.g. $e_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$, $e_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}$, $e_n = \begin{bmatrix} 0 \\ \vdots \\ 1 \end{bmatrix}$)

Kronecker delta

$\delta_{ij} :=$

Claim: $e_1, \dots, e_n$ is l.i. (generalizes $\underbrace{\begin{bmatrix} 1 \\ 0 \end{bmatrix}}_{e_1}, \underbrace{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}_{e_2}$ in $\mathbb{R}^2$)

Pf: $0 = \sum_{i=1}^n a_i e_i = \sum_{i=1}^n a_i \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix}$ $\leftarrow$ i th entry/row is 1
rest is 0

$$= \begin{bmatrix} a_1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ a_2 \\ 0 \\ \vdots \\ 0 \end{bmatrix} + \dots + \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ \vdots \\ a_n \end{bmatrix}$$

$$\Rightarrow a_1 = \dots = a_n = 0 \Rightarrow e_1, \dots, e_n \text{ l.i.}$$

□

(2) $1, x, x^2, \dots, x^n \in \mathbb{R}[x]_{\leq n}$ is l.i.

Pf: Exercise.

□

(3) MATH 161: $\sin x, \cos x \in \mathcal{F}(\mathbb{R}, \mathbb{R})$ is l.i.